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To cite this article: You Jin Jeong & Gahgene Gweon (2021) Advantages of Print Reading over Screen Reading: A Comparison of Visual Patterns, Reading Performance, and Reading Attitudes across Paper, Computers, and Tablets, International Journal of Human-Computer Interaction, 37:17, 1674-1684, DOI: [10.1080/10447318.2021.1908668](https://doi.org/10.1080/10447318.2021.1908668)

To link to this article: <https://doi.org/10.1080/10447318.2021.1908668>



Published online: 28 Apr 2021.



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Advantages of Print Reading over Screen Reading: A Comparison of Visual Patterns, Reading Performance, and Reading Attitudes across Paper, Computers, and Tablets

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ABSTRACT

We examined the effects of the reading medium (print vs. digital) on readers' visual patterns, reading performance, and reading attitudes. Two within-subject experiments were conducted with 74 readers, who read articles using three reading media: print, computer, and tablet. The experimental results showed that in terms of visual patterns, readers exhibited a shorter fixation duration and a higher fixation count during print reading than during screen reading; reading performance, as measured on the basis of reading comprehension and reading time, was equivalent across all three media; however, in terms of reading attitude, readers reported higher levels of perceived understanding, perceived confidence, and perceived immersion and lower levels of perceived fatigue for reading printed text than reading from a device screen. Therefore, the performance gap between print and screen reading is narrowing. However, printed text may still be the preferred mode of reading, as demonstrated by the readers' preferences.

1. Introduction

As reading on digital devices is becoming increasingly popular, traditional printed materials are being replaced by computer screens and tablets. Multiple studies have shown how print and screen reading differ in terms of readers' cognitive and behavioral aspects, such as eye movement (Köpper et al., 2016; Zambarbieri & Carniglia, 2012), reading comprehension (Singer & Alexander, 2017), and attitudes (Eijansantos et al., 2020). However, there are mixed results on whether reading from paper or a screen is preferred. Some studies reported increased reading time (Liu, 2005) and lower comprehension scores (Mangen et al., 2013; Støle et al., 2020) during screen reading than during print reading. However, other studies reported that reading comprehension and reading time are similar (Hermena et al., 2017; Sage et al., 2019; Singer & Alexander, 2017). Therefore, a consensus on these differences has not yet been reached.

One possible reason for this disagreement is that numerous factors affect the reading experience. Reading is a multi-faceted process influenced by readers' word recognition, reading strategies, comprehension, and motivation (McLaughlin, 2012); thus, different reading media have varying advantages. Accordingly, Dillon (1992) claimed that examining single- or few-factor differences between screen reading and print reading is insufficient. Keller (2012) suggested five aspects of reading that should be considered when comparing print and screen reading. These include physical and attitudinal factors, such as eye strain and feelings, and performance factors, such as reading comprehension. However, studies in which multiple aspects of reading are measured and compared are rare. As the transition from printed sources to electronic sources continues, a comprehensive study

in which the medium's effect on the reading experience is examined both timely and necessary. Therefore, we propose to examine how physical, performance, and psychological factors compare during print and screen reading in an experimental study.

In this experiment, three aspects were selected to investigate a reader's overall reading experience: visual patterns, reading performance, and reading attitudes. Visual patterns, which are measured by eye movements, are a well-researched physiological measure that can be used to infer the cognitive and perceptual workloads during reading (Rayner et al., 2010). Although a few studies have been conducted to compare print and screen reading and examine visual patterns (Kretzschmar et al., 2013; Siegenthaler et al., 2011; Zambarbieri & Carniglia, 2012), mixed results have been reported. For example, participants exhibited longer fixations when reading printed books over digital devices (Siegenthaler et al., 2011). However, in Zambarbieri and Carniglia (2012) study, fixation duration was longer on a PC than on an iPad, Kindle, or printed book. In another study, fixation duration measured during computer reading was longer than print, tablet, and smartphone reading (Miranda et al., 2018). Given these mixed results, additional studies would help clarify visual patterns during reading on different media.

The second aspect of examining the reading experience is reading performance, measured through comprehension and reading time. Comprehension is the reader's ability to process text and understand and integrate objective information (Johnston, 1983). Reading time is the amount of time that a participant needs to read a given text (Sage et al., 2019). Both measures are used to compare medium effects on the reading

experience, as examined in various meta-analyses (Clinton, 2019; Kong et al., 2018). Earlier studies reported that reading on electronic devices takes longer and yields lower comprehension scores than reading on paper. For example, Liu (2005) showed that, in screen reading, less time is spent on concentrated reading and the level of sustained attention is decreased. In another study involving children, it was shown that reading from electronic screens disturbs the formation of cognitive maps of text structures, which is reflected in poor comprehension and recall of the presented material (Kerr & Symons, 2006). However, studies conducted after 2010 have shown no significant difference in reading comprehension between print and digital presentations (Holzinger et al., 2011) and claim that learning differences between print and screen reading have diminished in recent years (Kong et al., 2018). For example, Sage et al. (2019) compared college students' reading of educational articles and demonstrated that students spent equivalent time and learned equally well from paper, computers, and tablets.

Finally, understanding readers' attitudes toward the reading materials provides a psychological assessment of reading conditions that might differ from reading outcomes, such as reading comprehension. Readers' attitudes toward the reading materials include assessments of beliefs, opinions, emotions, and evaluations of reading and practical reading behaviors (Lewis & Teale, 1982). From a cognitive science perspective, a person's perception or self-assessment (e.g., perceived levels of understanding or difficulty) does not always reflect objective measures of behavior, such as visual patterns and reading comprehension (Rayner et al., 2006). Therefore, in addition to measuring reading comprehension, measuring reading attitude by asking about perceived aspects of reading, such as difficulty, confidence, and preference, may yield different insights on the reading experience. In fact, studies have demonstrated that reading attitudes influence reading comprehension. For example, intrinsic motivation, perceived difficulty (Steensel et al., 2019), and self-assessed confidence in comprehension (Agler et al., 2019) have all been shown to impact reading achievement. In studies that compare reading attitudes across different reading media, readers perceived greater visual fatigue, less immersion, and engagement during screen reading than paper reading (Hou et al., 2017; Jeong, 2012; Mangen & Kuiken, 2014). In our study, we also examined how attitudinal measures compare with visual pattern and reading performance measures.

Given the importance of examining visual patterns, reading performance, and reading attitudes, we study these variables to gain insight into how print and screen reading experience compare considering these multidimensional aspects. As computers are often used for reading, many studies have focused on comparing paper and computer reading (Mangen et al., 2013). As handheld digital devices are becoming more popular, a recent meta-analysis emphasized the need for further research on handheld devices such as tablets (Delgado et al., 2018). In our investigation, we used computer and tablet devices with an LCD rather than electronic ink. Moreover, because individual differences can account for a preferred reading medium, we chose a within-subject design for the experiment. Recent meta-analyses have revealed that a between-subject design is more commonly used than a within-subjects design in research to compare print and screen reading (Delgado et al., 2018; Kong et al., 2018). The within-subject design enables individual variability to be considered when examining the reading experience across the media.

Two experiments were conducted in our study. The first experiment examined the differences in visual patterns and reading performance across paper, computer, and tablet readings. Based on the eye movement, survey results, and interview data analysis of this experiment, the second experiment measured psychological variables, such as the readers' evaluations of difficulty, understanding, confidence, immersion, and fatigue. The experiment also assessed their visual patterns and reading performance. The study contributions are summarized as follows:

- Readers exhibited higher cognitive loads during digital reading than print reading, as evidenced by visual patterns showing longer fixation durations and lower fixation counts.
- Unlike the differences in visual patterns, reading performance did not differ between print reading and digital reading, as measured by comprehension and reading time.
- Although there was no difference in reading performance, readers rated the print reading experience more positively than the digital reading experience, as indicated by higher levels of perceived understanding, perceived confidence, and perceived immersion and lower levels of perceived fatigue.

2. Materials and method

We conducted two experiments to investigate differences in readers' visual patterns, reading comprehension and attitude across paper, computer, and tablet. The first experimental design focused on measuring how well readers could perform reading tasks in different media settings. Readers' visual patterns and reading performance were investigated across paper, computer, and tablet reading. The visual patterns were measured by fixation duration and fixation count; reading performance was measured by reading time and comprehension.

However, after reviewing participants' feedback from experiment 1, we found that readers' attitudes during reading are important as well. Some researchers have recognized that the subjective perceptions of readers might not be reflected in the objective measures of reading activity, such as eye movements, reading speed, and reading comprehension (Kretschmar et al., 2013; Liversedge & Blythe, 2007; Rayner et al., 2006). Therefore, our second experiment was designed to examine readers' reading experience across print and digital media and examine attitudinal factors along with performance factors. Five measures of reader attitude were assessed: perceived difficulty, perceived understanding, perceived confidence, perceived fatigue, and perceived immersion.

2.1. Experimental setup

This section details the reading materials used for the reading task and the technical details of the eye-tracking device and reading platforms used in the two experiments. The reading materials were selected from a reading comprehension section of the Korean Broadcasting System's Korean Language Proficiency Test, which is a national language exam that assesses Korean linguistic understanding and ability. To ensure that all passages were similar, the materials were chosen according to three guidelines. First, all

selected passages were to be of similar length and readability; each text was 284–307 words long and composed of four paragraphs (paragraphs had 51–107 words). Second, the texts and questionnaires should have a similar difficulty level, so easy-to-intermediate comprehension level texts and questionnaires were selected. Third, because participants may have expertise in certain subjects, it was important to choose reading materials that represented various topics to minimize any potential influence from a participant's knowledge. The chosen reading texts covered nine different topics: philosophy, natural science, literature, economics, politics, engineering, the arts, biology, and history.

The eye-tracking device used for the experiment was Tobii Pro Glasses 2, which captures real-time user gaze data. Tobii Studio 1.3 and Tobii Pro Lab 1.73 software were employed to extract and analyze the eye-tracking data. As shown in Figure 1, the reading text was presented in three media formats: paper, computer, and tablet. The computer was a 21.5" Apple iMac with a Retina display monitor at a resolution of 1920×1080 pixels. The tablets were a Microsoft Surface Pro with PixelSense™ used in experiment 1. A 12.9" Apple iPad Pro with a Retina display monitor and a resolution of 2732×2048 pixels was used in experiment 2. Although the media formats differed, the reading material was designed to have the same page layouts regardless of the device: A4-size paper or 8.3" x 11.7" surface area, 10-point Nanum Myeongjo font, and 8-point line spacing.

2.2. Experimental protocols

Both experiments employed a within-subject experimental design in which each participant read short passages on paper, a computer, and a tablet. To adjust for potential carry-over effects, each participant received the reading media and reading materials in random order. The details of the two experimental protocols are described below.

Figure 2 shows the procedure for experiment 1. As participants arrived for the experiment, an instructor explained the objectives and process of the study. After signing an informed consent form, participants were asked to complete a pre-reading questionnaire and a reading comprehension (RC) test. Next, they were asked to wear the eye-tracking device shown in Figure 2. In the central process of the experiment, participants performed six reading tasks with a one-minute break between tasks. As the participants performed each task, their reading times were recorded. However, participants were not informed that they were being timed to reduce time pressure on performance. Participants read two

independent passages on each medium (paper, computer, and tablet). Following each reading task, participants were asked to answer questions to assess their reading comprehension. After completing all reading tasks and assessments, participants were debriefed and received 10 USD in compensation for their time.

Figure 3 shows the procedure for experiment 2. As in the first experiment, the participants were invited to a quiet room and were informed of the experiment's purpose and procedure before signing a consent form. Next, the participants completed a pre-reading survey questionnaire and an RC level assessment. The final phase of pre-processing included wearing the eye-tracker device. The experiment was divided into two sessions, and participants had a five-minute break between sessions. In the first session, the participants were asked to read three randomly assigned passages using the three different media. Following each reading task, participants were asked to answer questions designed to assess reading comprehension, and they were allotted a one-minute break between each reading task. In the second session, participants were asked to read six randomly assigned passages that were evenly distributed across the three different media. After finishing each reading task, participants were asked to take an RC test and complete a survey that measured their reading attitudes. After completing the second session, participants engaged in a five-minute interview with an instructor, which they were informed was being recorded. Finally, the participants were debriefed and paid 10 USD in compensation for their time.

2.3. Subjects

A total of 74 college students participated in our study. Details of the participants' backgrounds and daily reading behaviors are presented in Table A1. In experiment 1, 29 college readers (19 females and 10 males) participated. Participants had a mean age of 27 ($SD = 3.99$) years and majored in different subjects: 36.7% engineering, 24.1% applied science, 17.2% humanities, and 20.7% social science (see the Appendix). RC assessment scores demonstrated that participants across all majors had similar levels of reading comprehension ($M = 8.93$, $SD = 1.07$). For experiment 2, 45 undergraduate and graduate readers (29 females and 16 males) participated. Participants had a mean age of 26.3 years ($SD = 3.24$) and represented a wide variety of majors: 28.9% humanities, 15.6% social sciences, 33.3% applied sciences, and 22.2% engineering. All participants demonstrated similar levels of reading comprehension ($M = 8.89$, $SD = .86$).

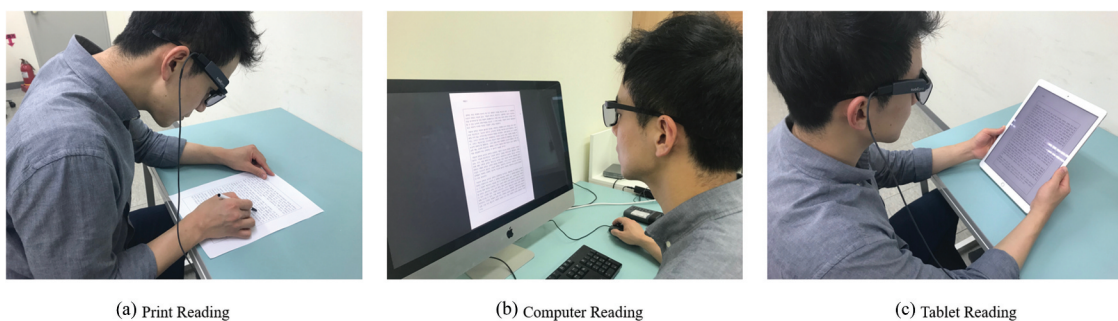


Figure 1. Three experimental conditions.

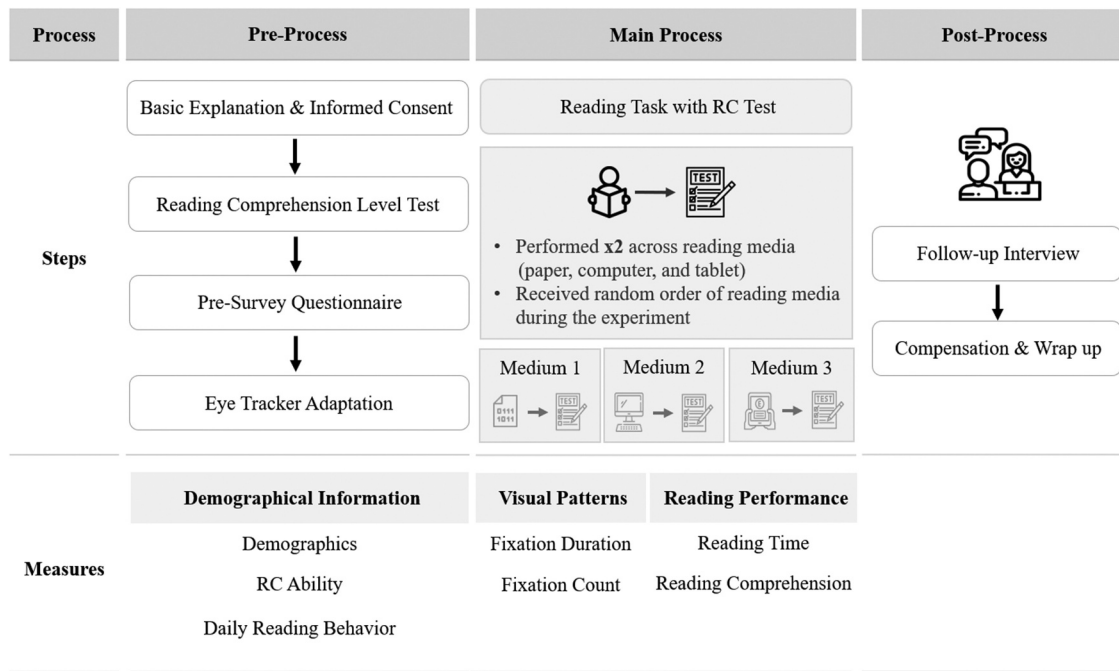


Figure 2. Procedure for experiment 1.

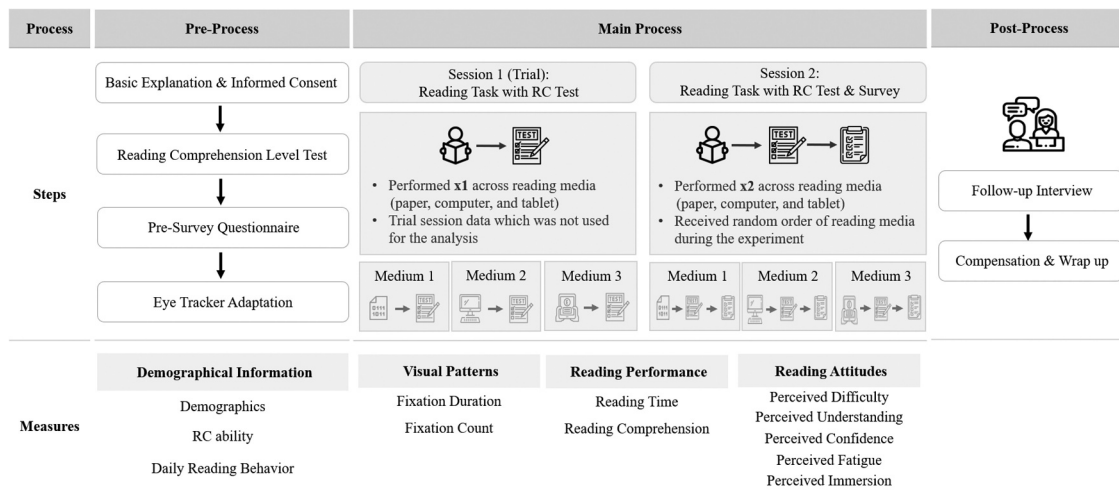


Figure 3. Procedure for experiment 2.

2.4. Measurements

In the two experiments, participants first completed a pre-reading survey consisting of 14 questions intended to measure nine variables: demographics, usage patterns of digital devices, medium preference, medium familiarity, medium utility, reading habits, reading performance, skepticism of screen reading, and self-perception of reading ability. Both experiments measured visual patterns using two eye movement measures (fixation duration and fixation count) and reading performance measures (reading time and reading comprehension). Five measures of reading attitudes (perceived difficulty, perceived understanding, perceived confidence, perceived fatigue, and perceived immersion) were also obtained in experiment 2.

2.4.1. Visual pattern measurements

The visual pattern was measured in terms of fixation duration and fixation count. Fixation duration was calculated by computing the average value of fixation durations measured while the participant read the entire text on a given page. For the fixation count, the total number of fixation points within the entire text of a given page was counted. Loading fixations above and below the cutoff ranges were filtered out; the cutoff range was from below 100 ms and above 500 ms. The cutoff ranges were determined by the standard deviation, specifically two SDs above and below the average.

2.4.2. Reading performance measurements

Reading performance was measured in terms of reading time and comprehension. Reading time was determined as the

number of seconds taken to complete each reading activity, beginning when the participant's eyes started to move until they stopped. All reading activities were recorded via the eye-tracking device, and the reading time was recorded and analyzed using the eye-tracking software. The data were verified by the researchers. To measure reading comprehension, participants were asked five questions for each text selection. These were true/false questions that assessed a participant's general understanding of the reading material. Each correct item scored one point, with a maximum score of five points. As there were two reading passages per medium, a maximum score of ten points was possible for each reading medium.

2.4.3. Reading attitude measurements

In experiment 2, we measured five perceived reading attitudes: difficulty, understanding, confidence, fatigue, and immersion. An explanation of how each reader's attitude measure was calculated is provided as follows.

First, perceived difficulty refers to the difficulty of each reading task as judged by the participant, who was asked "How difficult did you perceive this reading task to be?" A perceived difficulty score was calculated for each reading task, ranging from 1 (very easy) to 4 (very difficult).

Second, the levels of perceived understanding were measured by asking, "How clearly did you understand the reading text?" The scale ranged from 1 (poorly understood) to 4 (completely understood) and was used to measure how participants assessed their comprehension of the assigned material.

Third, the levels of perceived confidence measure how well the participants think they scored on their RC tests for each reading task. This was measured by asking, "How confident are you about your reading comprehension?" The scale ranged from 1 (not confident at all) to 4 (fully confident).

Fourth, the levels of perceived fatigue were measured using two questions derived from the literature (Hou et al., 2017): "My eyes became tired as I read the text" and "I felt comfortable reading the text (reverse-scored)." The Likert scale, which ranges from 1 (very unlikely) to 4 (very likely), was used to measure the visual fatigue due to participants' reading experiences.

Finally, the levels of perceived immersion were measured using two questions derived from the literature (Hou et al., 2017): "I became really involved in reading the text" and "I could not focus on reading the text (reverse-scored)." The Likert scale, which ranges from 1 (strongly disagree) to 4 (strongly agree), was used to measure whether participants felt engaged while reading.

2.5. Data analysis

The Statistical Package for the Social Sciences (SPSS) Version 23 was used to examine the statistical difference between two visual patterns and two reading performance measures across print, computer, and tablet readings. Repeated measurements with ANOVA (rmANOVA) with a significance threshold of $p < .05$ was used to determine how the three independent variables (print, computer, and tablet) differed in response to the measured dependent variables (fixation duration, fixation count, reading time, reading comprehension, perceived difficulty, perceived understanding, perceived confidence, perceived fatigue, and perceived immersion). All analyses were corrected using the Bonferroni post-hoc test.

3. Result

This section presents a statistical analysis of the data obtained from the two experiments. The two subsections provide the results from experiment 1 and experiment 2 regarding our hypothesis on visual patterns, reading performance, and reading attitude.

3.1. Experiment 1

Our analysis of visual patterns using the rmANOVA test indicated that the fixation duration ($p = .054$) and fixation count ($p = .055$) were statistically different across print, computer, and tablet readings, as shown in Table 1. This experiment also evaluated our hypothesis of reading performance: that readers' reading time and comprehension would differ across print, computer, and tablet readings. The results of the rmANOVA in Table 1 demonstrate that unlike our hypothesis, the reading time (Wilk's Lambda = 0.92, $F(2, 22) = 0.98$, and $p = .393$) and reading comprehension (Wilk's Lambda = 0.99, $F(2, 25) = 0.04$, and $p = .965$) did not differ significantly across the three media.

Table 2 presents the results of the Bonferroni post-hoc test, which was used to demonstrate how readers' visual patterns differed across the three media. Specifically, a larger difference in fixation duration was observed between print and computer readings ($p = .046$) than between print and tablet readings ($p = .028$). A significant difference in the fixation count was observed only between print and tablet readings, with a higher fixation count on print reading than on tablet reading ($p = .040$).

Table 1. Experiment 1: repeated ANOVA for visual patterns and reading performance measurements across print reading (PR), computer reading (CR), and tablet reading (TR).

Variable	Measure (unit)	PR mean (SD)	CR mean (SD)	TR mean (SD)	<i>F</i> (df)	Sig.
Visual patterns	Fixation duration (ms)	234.93 (11.82)	283.12 (21.36)	275.38 (17.30)	3.70 (1.30)	.054*
	Fixation count (count)	4929.86 (416.12)	4760.82 (470.50)	4344.78 (359.60)	2.46 (1.80)	.055*
	Reading time (s)	136.29 (9.63)	128.73 (11.15)	125.18 (8.66)	.98 (2.00)	.393
Reading performance	Reading comprehension (0–10 point scale)	8.59 (0.22)	8.52 (0.20)	8.59 (0.28)	.04 (2.00)	.965

* $p \leq 0.05$.

Table 2. Experiment 1: post-hoc comparison of visual patterns measurements between print reading (PR), computer reading (CR), and tablet reading (TR).

Variable	Measure	PR – CR		PR – TR		CR – TR	
		Diff	Sig.	diff	Sig.	diff	Sig.
Visual patterns	Fixation duration (ms)	–39.20*	.046	–31.45*	.028	7.74	.673
	Fixation count (count)	169.04	.590	585.10*	.040	416.04	.080

* $p \leq 0.05$.

3.2. Experiment 2

Table 3 displays the results of the rmANOVA for visual patterns, reading performance, and reading attitudes across print, computer, and tablet readings. Table 4 presents the results of the Bonferroni post-hoc test, which showed a significant difference between readers' visual patterns and reading attitudes across the three types of media. For visual patterns, the effect of media type on fixation duration ($p = .001$) and fixation count ($p = .019$) was significant, as seen in Table 3. In the rmANOVA, the differences in reading performance, measured in terms of the reading time (Wilk's Lambda = 0.92, $F(2, 43) = 0.98$, and $p = .393$) and reading comprehension (Wilk's Lambda = 0.99, $F(2, 43) = 0.11$, and $p = .898$), mirrored those of the first experiment and were statistically insignificant across media types. When measuring participants' reading attitudes, four out of the five measures were statistically significant across the media, as shown in Table 3 [perceived understanding ($p = .019$), confidence ($p = .019$), perceived fatigue ($p = .000$), and perceived immersion ($p = .009$)]. The perceived difficulty of the reading was the only measure that showed no statistically significant difference across print, computer, and tablet readings ($p = .583$).

As shown in Table 4, the Bonferroni post-hoc test indicated that a significant difference in fixation duration was observed only between print and computer readings ($p = .000$). It is important to note that a fixation duration difference was also observed between print and tablet readings in experiment 1. A statistically significant difference in readers' fixation counts was observed only between print and tablet readings ($p = .030$), which is consistent with the data from the first experiment. A statistically significant difference in reading attitudes was observed in perceived understanding ($p = .015$) and perceived confidence ($p = .014$) between print and tablet reading. For the perceived fatigue and perceived immersion measures, significant differences were observed between print and computer reading (PR-CR) and print and tablet reading (PR-TR).

4. Discussion

This study was designed to investigate the effects of selected media on the reading experience for three variables (visual patterns, reading performance, and reading attitudes) across

Table 3. Experiment 2: repeated ANOVA for visual patterns, reading performance, and reading attitudes across print reading (PR), computer reading (CR), and tablet reading (TR).

Variable	Measure (unit)	PR mean (SD)	CR mean (SD)	TR mean (SD)	$F(df)$	Sig.
Visual patterns	Fixation duration (ms)	234.97 (10.85)	288.18 (15.33)	259.71 (11.70)	9.02 (1.77)	.001***
	Fixation count (count)	4604.03 (195.03)	4575.36 (215.23)	4178.56 (183.35)	4.34 (2.00)	.019**
Reading performance	Reading time (s)	134.65 (4.76)	127.62 (5.32)	126.91 (4.46)	.98 (2.00)	.393
	Reading comprehension (0–10 point scale)	8.04 (0.18)	8.13 (0.17)	8.16 (0.22)	.108 (2.00)	.898
Reading attitudes	Perceived difficulty (0–4 point scale)	2.50 (0.07)	2.60 (0.07)	2.58 (0.07)	.55 (2.00)	.583
	Perceived understanding (0–4 point scale)	2.70 (0.07)	2.58 (0.06)	2.44 (0.07)	4.34 (2.00)	.019**
	Perceived confidence (0–4 point scale)	2.60 (0.08)	2.47 (0.06)	2.36 (0.08)	4.38 (2.00)	.019**
	Perceived fatigue (0–4 point scale)	1.96 (0.05)	2.39 (0.08)	2.49 (0.07)	20.01 (2.00)	.000**
	Perceived immersion (0–4 point scale)	2.96 (0.07)	2.76 (0.007)	2.72 (0.08)	5.39 (1.71)	.009**

* $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$.**Table 4.** Experiment 1: post-hoc comparison of visual patterns and reading attitudes measurements between print reading (PR), computer reading (CR), and tablet reading (TR).

Variable	Measure	PR – CR		PR – TR		CR – TR	
		diff	Sig.	diff	Sig.	diff	Sig.
Visual patterns	Fixation duration (ms)	–53.21***	.000	–24.75	.123	28.46	.174
	Fixation count (count)	28.68	1.000	425.47*	.030	396.79	.059
Reading attitudes	Perceived understanding (0–4 point scale)	.12	.463	.26**	.015	.13	.269
	Perceived confidence (0–4 point scale)	.13	.368	.24**	.014	.11	.074
	Perceived fatigue (0–4 point scale)	–.44***	.000	–.52***	.000	–.08	.750
	Perceived immersion (0–4 point scale)	.21*	.012	.24*	.042	.03	.073

* $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$.

Table 5. Summary of visual patterns, reading performance, and reading attitudes differences across print reading (PR), computer reading (CR), and tablet reading (TR).

Variable	Measure	Experiment 1 (N = 29)	Experiment 2 (N = 45)
Visual patterns	Fixation duration	PR < TR* < CR*	PR < TR < CR*
	Fixation count	PR > CR > TR*	PR > CR > TR*
Reading performance	Reading time	PR = CR = TR	PR = CR = TR
	Reading comprehension	PR = CR = TR	PR = CR = TR
Reading attitudes	Perceived difficulty	-	PR = CR = TR
	Perceived understanding	-	PR > CR > TR*
	Perceived confidence	-	PR > CR > TR*
	Perceived fatigue	-	PR > CR* > TR*
	Perceived immersion	-	PR > CR* > TR*

* indicates a statistically significant difference.

print, computers, and tablets. Table 5 summarizes the results from both experiments. This section discusses the insights gained from the experimental data analysis.

4.1. Screen readings' visual patterns imply higher cognitive loads and frequent skimming

Eye-movement analysis was used in this study to quantify differences in readers' fixation durations and counts across the three media types to examine how the medium affects cognitive processing and attention. Our results are discussed in terms of the characteristics of digital devices and reading patterns. First, the characteristics of digital devices can impact eye movement. Some literature has suggested that some characteristics of digital media, such as screen contrast, optical strain, and display quality, increase eye fatigue and distract readers from concentrated reading by requiring more cognitive effort to process information (Baron, 2015; Hou et al., 2017; Mayes et al., 2001; Wästlund & Henrik Reinikka, 2005). Our data showed that reading on digital devices caused users to exhibit longer fixation durations, indicating higher cognitive loads and effort (Just & Carpenter, 1980; Shojaeizadeh et al., 2016). However, when reading on digital devices, users in our study had a lower fixation count. This relates to the second point of our discussion. Second, screen reading is associated with F-shaped reading patterns, which has been reported more frequently on digital media than on paper (Biedert et al., 2012; Chaparro et al., 2007). The F-shaped reading pattern is a reading pattern that is characterized by fixation concentration at the top and left side of the page (Nielsen, 2006). Engaging in F-shape reading pattern is associated with skimming and paying less attention to text. We suspect that the lower fixation count in our data may have occurred in digital reading compared to print reading because of the F-shaped reading pattern's higher occurrence. Liu (2005) suggested that younger generations are digital natives who employ nonlinear reading and skimming more often; this competes with the traditional ways of close, or linear, reading.

4.2. Reading performance is equivalent across all media

Readers performed equally well on reading time and comprehension across all three media ($p > .05$). These results are consistent with recent studies reporting a narrowing gap in

reading time and reading comprehension between print and screen readings (Eden & Eshet-Alkalai, 2013; Kurata et al., 2016; Singer & Alexander, 2017; Zambardi & Carniglia, 2012). Readers' familiarity with digital devices and high digital literacy may explain this narrowing gap. First, the participants of this study were young, digital-native college readers in South Korea who were familiar with digital devices. It is possible that the participants' frequent screen reading experience could have eliminated differences in performance between print and screen because any increase in medium familiarity affects learning positively (Chen et al., 2014; Sage et al., 2019). Second, the high literacy levels of the participants may have created a ceiling effect. Despite the varied readings covering the humanities, politics, and natural science, the average participant RC score was 8.91 out of 10. Therefore, age and literacy levels may be important moderating variables in understanding print and screen reading differences that can affect the reading experience on paper and digital devices. A more diverse population should be investigated in future studies to evaluate different cultures and geographical areas.

4.3. Readers prefer print over screens and computers over tablets

An interesting finding in this study was that, despite equal reading performance across all media, readers' perceived understanding and perceived confidence levels were lower on digital devices than on print ($p < .01$). Past research has established a positive relationship between performance and perceptions (Kettanurak Vichuda & Haseman, 2001; Sage et al., 2019). However, the data in our study suggest that reading performance does not necessarily correlate with readers' perceived reading performance. In addition, there is no evidence of "screen inferiority" in the data, which refers to screen reading that is associated with lower performance and overconfidence in reading comprehension (Ackerman & Goldsmith, 2011). As the use of screen reading continues to increase, investigating the perpetual relationship between readers' actual and perceived reading performance on digital devices could prove interesting. The results show that medium preference and confidence in reading ability play important roles in readers' actual and perceived reading performance. Future studies should investigate how readers' performance varies with technology adaptation, reader medium preference, and reader confidence.

Readers showed more positive attitudes and stronger preferences for paper than digital devices (Table A1). They reported greater fatigue and lower levels of immersion while reading on computers and tablets ($p < .01$). Fatigue imposes an extra load on the cognitive processing system. It may prevent readers from forming a cognitive map of information, so readers may feel less engaged on digital devices and do not understand the text as well (Hou et al., 2017). In addition, the survey of daily reading behaviors indicated that paper is still a preferred medium and has advantages over screens. The two experiments showed that 48.6% of participants use paper in their daily reading, and 70.2% prefer paper in daily use. Furthermore, 81.0% selectively use media depending on the reading materials. Readers' preferences of medium depended on what they were reading; they preferred paper for reading academic or educational materials and digital devices for reading short, entertaining content such as news, blogs, comics,

and social media. Even though screen reading is more commonly used in education and viewed more positively than ever before, readers still prefer printed text when they require an in-depth understanding of the material.

In this study, we also compared readers' experience on print versus digital devices and computers versus tablets. Because of tablets' portability, some researchers have suggested that tablets used in educational settings will create a reader-centered environment and increase learning engagement (Fischer et al., 2013; Rikala et al., 2013). However, the results demonstrate that tablets might not be a better choice for reading than computers; tablets scored the lowest for all reading attitude measures (Table A1). This could be associated with medium familiarity, as Chen et al. (2014) have suggested. Medium familiarity could be a potential influence on reading on digital devices. Participants with high levels of familiarity performed significantly better on comprehension than participants with low levels of medium familiarity. Accordingly, participants in this study reported higher familiarity with computers than tablets (Table A1). Although most participants had prior reading experience with paper, computers, and tablets, only 10.81% of participants ($n = 8$) reported using tablets as their daily reading device, and 5.4% of participants ($n = 4$) chose the tablet as their most familiar reading medium. In addition to these data, participants' post-reading interview statements from experiment 2 affirmed the potential impact of medium familiarity on the reading experience. Although most participants acknowledged that the tablet provides a paper-like reading experience in terms of sight angle, they also said that it was the least comfortable to use in terms of medium familiarity and characteristics that inhibited focused reading. The data highlighting the unfamiliarity of tablet use shows that medium familiarity is an essential part of understanding reading activity across different reading media. Consequently, future studies should investigate the effect of medium familiarity on readers' eye movements.

5. Conclusion

As readers have become increasingly familiar with screen reading due to the proliferation of digital devices, understanding the impact of the medium on reading activity also increases. This study revealed three valuable insights on how different media affect the overall reading experience: (1) regarding visual patterns, readers exhibited a shorter fixation duration and a higher fixation count for print reading than for screen reading; (2) reading performance, as measured by reading comprehension and reading time, were equivalent across all three media; and (3) regarding reading attitudes, readers reported higher levels of perceived understanding, perceived confidence, and perceived immersion, and lower levels of perceived fatigue for print reading than for screen reading. Based on these insights, it was concluded that readers' performances were not negatively affected by the medium, possibly due to their familiarity with digital devices. However, providing readers with printed text might still be best in a learning environment because readers have more positive reading attitudes toward print than digital devices.

This study is crucial for advancing the existing literature for several reasons. First, the digital devices (computers and tablets) selected for use in this study are commonly used for reading, and

the insights gained explain how they differ from paper. Previous studies have focused on reading e-resources from a computer and e-book environment. In contrast, this study was focused on comparing printed matter to text on a tablet or computer, which are increasingly used for reading, including educational purposes (Jones & Brown, 2011; Siegenthaler et al., 2011). In addition, these data further strengthen previous research by observing readers' visual patterns across different media. The results of the eye-movement analysis strengthen the argument that screen reading is related to higher cognitive loading and skimming, which may prevent the readers' ability to read in depth (Destefano & Lefevre, 2007; Hillesund, 2010). The eye-movement analysis also provides a meaningful contribution to the existing literature because limited research has been conducted to investigate the effect of print and screen reading on readers' eye movements.

Although this study provides valuable insights to previous studies, there are several limitations. First, this study was conducted with a sample of college readers in South Korea, which may not readily translate to other cultures or geographical locations. Future studies exploring this matter in diverse cultural contexts would be challenging but interesting. Second, smartphones were not included as a screen reading medium. According to a preliminary survey, readers reported that they were most likely to use smartphones for their daily reading. Because portability is an important determiner in device selection, future studies should examine the effects of smartphones on the reading experience and how they may differ from other digital devices in terms of visual patterns, reading performance, and reading attitudes. Finally, as visual patterns (measured by eye movements) could be influenced by the differences between paper, computers, and tablets, future studies should take note of the factors that can influence eye movements, such as variations in text and font, letter spacing, and line length (Jacobson & Dodwell, 1979; Kolars et al., 1981). In addition, because screen reading in our study was performed on an LCD device, further research is needed to verify whether an alternative type of display, such as electronic ink, would yield similar results for visual pattern, reading performance, and reading attitude.

Disclosure of potential conflict of interest

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the National Research Foundation of Korea (NRF) under Grant number 2017R1D1A1B03034511.

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Appendix

Table A1. Descriptive statistics of participants' demographics and daily reading behaviors.

Variables		Experiment 1 (N = 29)		Experiment 2 (N = 45)	
		n	%	n	%
Background					
Gender	Female	19	65.5	29	64.4
	Male	10	34.5	16	35.6
College major	Humanity	5	17.2	13	28.9
	Social science	6	20.7	7	15.6
	Applied science	7	24.1	15	33.3
	Engineering	11	36.7	10	22.2
Daily reading media & reading behavior					
Different use of media depends on reading material	Yes	23	79.3	37	82.2
	No	6	20.7	8	17.8
Daily reading media for main usage	Paper	8	27.6	23	51.1
	Computer	17	58.6	14	31.1
	Tablet	0	0.0	8	17.8
	All	4	13.8	-	-
Daily reading digital contents	News	14	48.3	7	15.6
	Academic or Learning	2	6.9	2	4.4
	Entertainment (blog, webtoon, etc.)	6	20.7	8	17.8
	e-book	2	6.9	-	-
	Social media	5	17.2	28	62.2
Most preferred media for daily reading activities	Paper	15	51.7	37	82.2
	Computer or laptop	8	27.6	5	11.1
	Tablet	1	3.4	3	6.7
	Neither	5	17.2	-	-
Most familiar media for daily reading activities	Paper	13	55.2	38	84.4
	Computer or laptop	7	27.6	5	11.1
	Tablet	2	6.9	2	4.4
	Neither	3	10.3	-	-
Media with the highest utility for daily reading activities	Paper	13	44.8	31	68.9
	Computer or laptop	5	17.2	6	13.3
	Tablet	8	27.6	8	17.8
	Neither	4	10.3	-	-